Question 1

a. The order form requires the information of all the included elements in the slip given. The part number is a natural PK for parts.

b.

partNumber	name	type	cageCode	quantityOrdered	unitPrice	**date**	**time**	employee	customerName	**customerNumber**	customerType
10654	Float Control	Plumbing	g413	4	12	7/1/2024	10:30	D.Harrision	Jeff Peterson	HG54587	consumer

Above shows 1NF the PK, without adding any additional fields, is a composite of the date, time, and customer number. There are no repeating categories. However, we have not reached 2NF because some elements above repeat in rows and there is some partial dependencies (customerName and type on customerNumber, all of the part fields on partNumber).

partNumber	quantityOrdered	**date**	**time**	***customerNumber***	employee
10654	4	7/1/2024	10:30	HG54587	D.Harrison

customerNumber	customerFirstName	customerLastName	customerType
HG54587	Jeff	Peterson	consumer

purchases

includes

partNumber	partName	type	cageCode	unitPrice
10654	Float Control	Plumbing	g413	12

The above shows 2NF. The tables are seperated so that there are no partial dependencies anymore. This also makes it 3NF as there are no transitive dependencies either.

c. see above

d. see above

Question 2

a. branchNo is a PK for the different branches of the institution. patNo is PK for the different patients, same with staffNo for staff. It seems that different patients go to see different therapists so it is a many to many relationship.

b.

staffNo	therapistName	patNo	patName	appointment date	appointment time	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

Currently, the table is not in 1NF yet because appointment is not atomized.

staffNo	therapistName	**patNo**	patName	**apptDate**	**apptTime**	branchNo
s1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15

Above is the table in 1NF form, without adding fields, there is a composite PK of the staffNo, patNo, apptDate, and apptTime. There are no repeated categories. However, we have not reached 2NF as there are still partial dependencies (patName on patNo, therapistName on staffNo).

staffNo	**patNo**	**apptDate**	**apptTime**	branchNo
s1011	P100	9/12/2022	10:00	M15

attends

staffNo	therapistName
s1011	Fred Smith

attends

patNo	patName
P100	Lily White

Seperating the tables now gets rid of those dependencies to bring it to 2NF, and because there are no transitive dependencies either, it is 3NF as well.

c. see above

d. see above

Question 3

a. All of the fields with No in it are primary keys. based on this table it seems that many employees can work at the same event with the same contrac No, but work different hours.

b.

eNo	contractNo	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

Currently, the table is in 1NF as there is a PK in the form of contractNo combined with eNo. However, because of partial dependencies (eventLoc only depending on eventNo, eName on eNo), it is not in 2NF.

contractNo	hours	***eNo***	***eventNo***
C1024	16	1135	H25

takes place at

requires

eventNo	eventLoc
H25	Queens

eNo	eName
1135	Smith J

Seperating the tables now gets rid of those dependencies to bring it to 2NF, and because there are no transitive dependencies either, it is 3NF as well.

c. see above

d. see above